MIL-D-45676D(MU)
31 August 1970
SUPERSEDING
MIL-D-45676C(MU)
14 October 1965

MILITARY SPECIFICATION

DRIVE, BALLISTICS: M10 SERIES

1. SCOPE

1.1 This specification covers several ballistics drives, each of which is included in the fire control system of its respective tank for the purpose of transmitting superelevation data determined by a ballistics computer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8205606	Drive, Ballistics: M10B1
F8620100	Drive, Ballistics: M10
F8620155	Drive, Ballistics: M10A1
F8620165	Drive, Ballistics: M10A2
F8620202	Drive, Ballistics: M10A3

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F8620350	Drive, Ballistics:	M10A4
F8621333	Drive, Ballistics:	M10A5
F10541900	Drive, Ballistics:	M10A6

PACKAGING DATA SHEETS

8205606	Packaging of Drive, Ballistics: M10B1
8620100	Packaging of Drive, Ballistics: M10
8620155	Packaging of Drive, Ballistics: M10A1
8620165	Packaging of Drive, Ballistics. M10A2
8620202	Packaging of Drive, Ballistics: M10A3
8620350	Packaging of Drive, Ballistics. M10A4
8621333	Packaging of Drive, Ballistics: M10A5
10541900	Packaging of Drive, Ballistics: M10A6

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The ballistics drive(s) specified in the contract or order shall be manufactured in accordance with the applicable drawing(s) listed in Section 2 and all drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connections

3.3 Environmental service conditions.

3.3.1 Standard ambient temperature.- The ballistics drive shall meet the requirements of 3.3.3 to 3.4.3.2 inclusive at a standard ambient temperature between +60° and +90°Fahrenheit (F).

3.3.2 Extreme operating temperatures.- The ballistics drive shall meet the requirements of 3.4.4.1 and 3.4.4.2 while exposed to, and thermally stabilized at, ambient temperatures of -40° and +150°F.

3.3.3 Vibration.- When mounted and secured in its normal operating position, the ballistics drive shall withstand a vibration of 30 ± 5 cycles per second (cps) frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration the drive shall show no evidence of damage or physical

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failure, and it shall meet all the performance requirements of this specification.

3.4 Performance.- When mounted in the normal operating position (Figure 1 or 2) the ballistics drive shall meet the requirements of 3.4.1 to 3.4.4.2 inclusive.

3.4.1 Superelevation.

3.4.1.1 Input travel.- The input shaft shall be rotatable clockwise and counterclockwise. The limit of travel shall be effected wholly by the stops on the split nut and extremes of the lead screw. There shall be no interference, jamming, or torque build-up as a result of approaching or reaching the stops. A minimum of 30 full turns of the input shaft shall be available over the entire range of rotation.

3.4.1.2 Accuracy.- With the 11-inch arm in a fixed position; with the qualification plate set for zero degrees horizontal, backlash excluded; with the 200 inch pounds load applied to the assembly (see Figure 1 or 2); and with the "0" graduation lines on the inner and outer scales of the double eccentrics in coincidence with their respective index lines, the ballistics drive shall meet the superelevation accuracy specified below:

Input Shaft Rotation* (Complete Revolutions)	Output Coupling Rotation	Tolerance
5 - clockwise	24.95 mils	±0.10 mil
10 - clockwise	49.94 mils	±0.10 mil

(*) Origin of input shaft rotation at qualification plate.

3.4.1.2.1 Backlash.- With no load applied to the assembly, backlash in the superelevation mechanism shall not exceed 0.1 mil as a result of change in direction of rotation of the superelevation input shaft. Backlash shall be checked at the following positions of the superelevation input shaft:

(a) Zero superelevation.
(b) Five (5) complete clockwise revolutions of the input shaft from the zero superelevation setting.
(c) Ten (10) complete clockwise revolutions of the input shaft from the zero superelevation setting.

3.4.2 Synchronization.- Synchronization, with the 200 inch pounds load applied (see Figure 1 or 2), shall cause an identical agreement in movement between the main gun centerline and the gunner's periscope line of sight.

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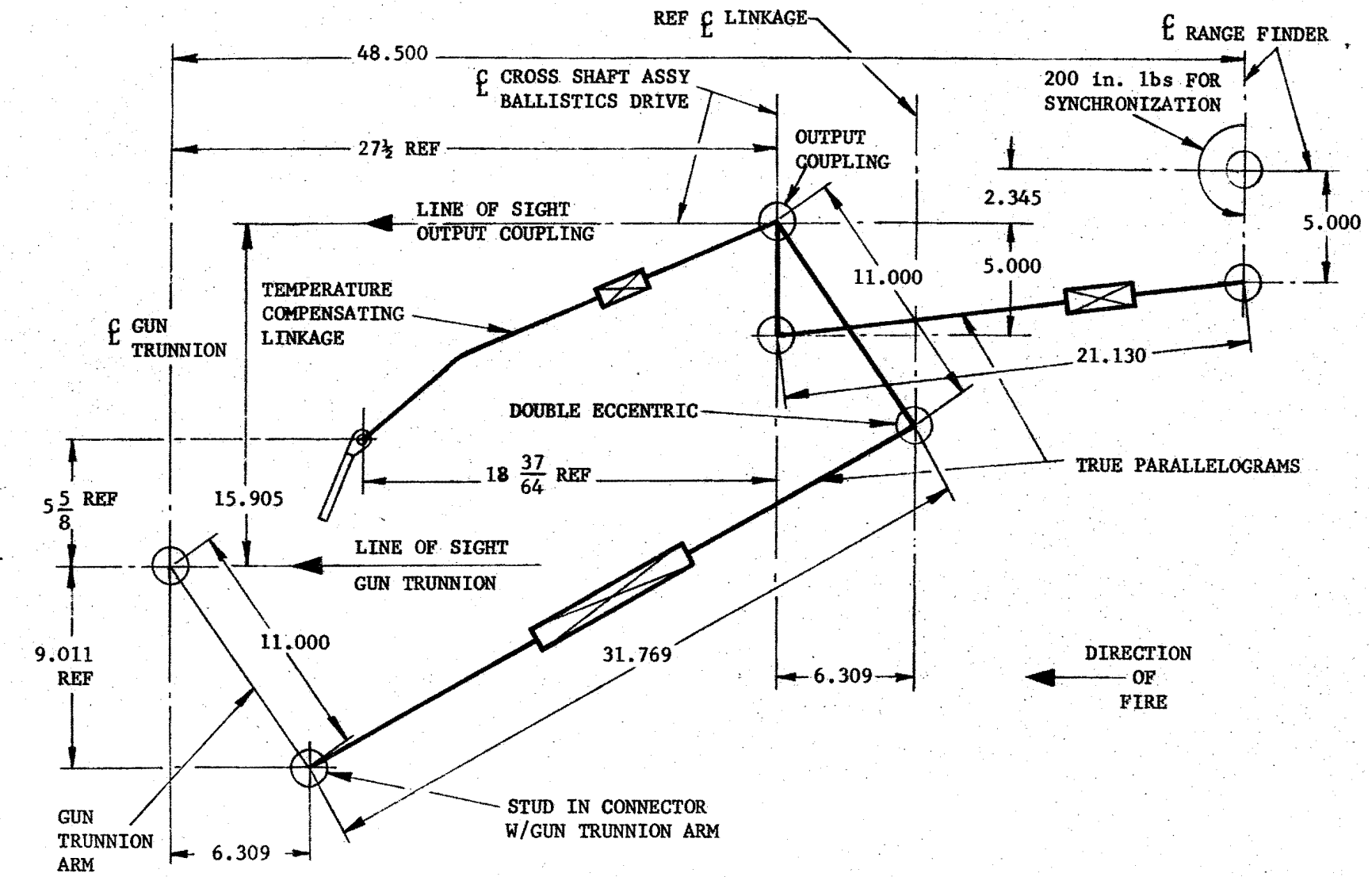


FIGURE 1

NOMINAL POSITIONING OF BALLISTICS DRIVES
M10, M10A1, M10A2, M10A6, and M10B1 ASSEMBLED IN TURRET

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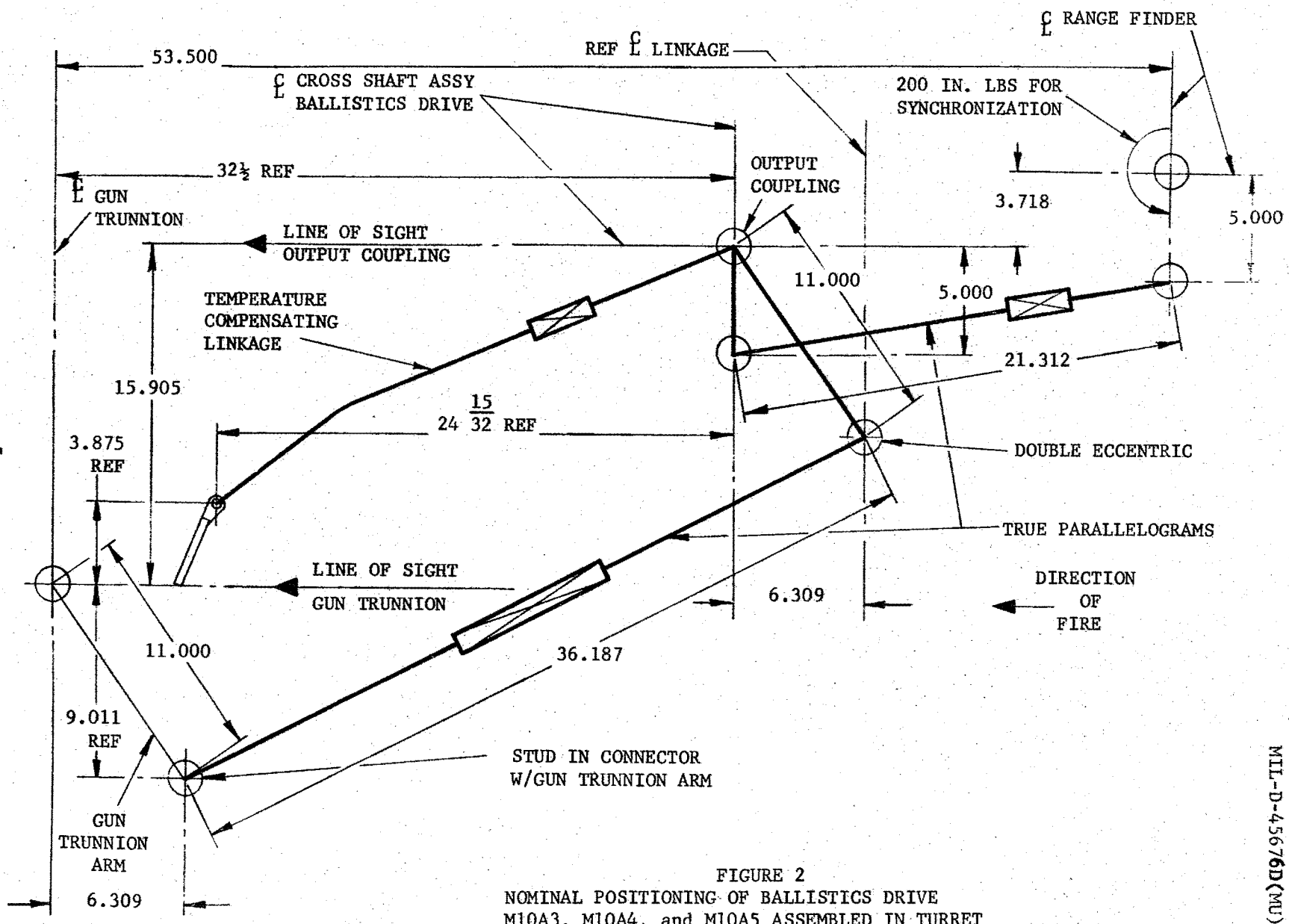


FIGURE 2
NOMINAL POSITIONING OF BALLISTICS DRIVE
M10A3, M10A4, and M10A5 ASSEMBLED IN TURRET

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3.4.2.1 Conditions for synchronization.- The ballistics drive must cause synchronization between the main gun centerline and the gunner's periscope line of sight within 0.2 mil when the conditions noted in 3.4.2.1.1 to 3.4.2.1.3, inclusive, are imposed.

3.4.2.1.1 A constant load of 200 inch pounds applied as specified on the following drawings.

D8270945 (Sheet No. 1) and Figure 1 herein-Ballistics Drive M10B1
D8620095 (Sheet No. 4) and Figure 1 herein-Ballistics Drive M10
D8620156 (Sheet No. 4) and Figure 1 herein-Ballistics Drive M10A1
D8620166 (Sheet No. 1) and Figure 1 herein-Ballistics Drive M10A2
D8620211 (Sheet No. 1) and Figure 2 herein-Ballistics Drive M10A3
D8620192 (Sheet No. 1) and Figure 2 herein-Ballistics Drive M10A4
D8620211 (Sheet No. 1) and Figure 2 herein-Ballistics Drive M10A5
D8620166 (Sheet No. 1) and Figure 1 herein-Ballistics Drive M10A6

3.4.2.1.2 All planetary error due to superelevation input shaft rotation excluded.

3.4.2.1.3 Movement in elevation and depression as follows:

(a) Elevation from -5 degrees to 0 degrees.
(b) Elevation from 0 degrees to +17 degrees.
(c) Depression from +17 degrees to 0 degrees.
(d) Depression from 0 degrees to -5 degrees.

3.4.2.2 Backlash.- Backlash at each of the above simulated gun positions shall not exceed 0.1 mil.

3.4.3 Operability (at standard ambient temperature).

3.4.3.1 Rotating studs.- The torque required to rotate the studs located at the free end of the linkages which connect to the simulated gun trunnion and rangefinder shall be 3.0 ± 0.5 pound inches.

3.4.3.2 Superelevation input shaft.- The torque required to rotate the superelevation input shaft (at its point of origin) shall not exceed 4 pound inches throughout the entire range of travel clockwise and counterclockwise.

3.4.4 Operability (at -40° and +150°F).

3.4.4.1 Rotating studs.- The torque required to rotate the studs identified in 3.4.3.1 shall not exceed the value obtained in 3.4.3.1 by more than 1 pound inch.

3.4.4.2 Superelevation input shaft.- The torque required to rotate the superelevation input shaft shall not exceed the value obtained in 3.4.3.2 by more than 1 pound inch.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 ballistics drives plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0% Defective			
101.	Vibration	3.3.3	4.6.2
102.	Superelevation, input travel	3.4.1.1	4.6.3.1
103.	Superelevation, accuracy	3.4.1.2	4.6.3.2
104.	Superelevation, backlash	3.4.1.2.1	4.6.3.3
105.	Synchronization	3.4.2	4.6.3.4
106.	Synchronization, backlash	3.4.2.2	4.6.3.5
107.	Operability, rotating studs	3.4.3.1	4.6.4.1
108.	Operability, superelevation input shaft	3.4.3.2	4.6.4.2
MINOR: NONE			

NOTE: The test in Table I shall be conducted at a temperature between +60° and +9-°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One ballistics drive shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

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TABLE II

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three ballistics drives shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
303.	Operability, rotating studs (-40° and +150°F)	3.3.2, 3.4.4.1	4.6.1, 4.6.5.1
304.	Operability, superelevation input shaft (-40° and +150° F)	3.3.2, 3.4.4.2	4.6.1, 4.6.5.2

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor, and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection

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procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Ballistics Drives M10, M10A1, M10A2, M10B1 and M10A6.- Special testing equipment utilized for these tests shall conform to Drawing F8227525, Fixture, Final Inspection for Drive, Ballistics.

4.5.2.3.2 Ballistics Drives M10A3, M10A4 and M10A5.- Special testing equipment utilized for these tests shall conform to Drawing F8636700, Test Equipment for Drives, Ballistics.

4.6 Test methods and procedures.

4.6.1 Extreme operating temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.2. The ballistics drive shall be exposed to each temperature extreme specified in 3.3.2 for a period of five hours. At the end of each five hour period, and while still at the temperature extreme, the ballistics drive shall be subjected to the tests specified in 4.6.5.1 and 4.6.5.2. The ballistics drive shall conform to the operability requirements of 3.4.4.1 and 3.4.4.2 for each temperature extreme specified in 3.3.2.

4.6.2 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.3.3 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The vibrating equipment shall be equipped with a suitable adapter to permit mounting and securing the ballistics drive in its normal operating position. The input and output shafts shall be immobilized. The ballistics drive shall be mounted on the vibrating equipment specified and vibrated in a vertical plane for a period of five minutes ± 15 seconds. Subsequent to vibration, the ballistics drive shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.3.3, prior to being subjected to the remaining tests specified herein. A visual examination of the ballistics drive shall be made for loose, missing, cracked, or otherwise damaged parts. A tactile examination shall be conducted to check functioning of moving parts for undue irregularities of movement, friction, or looseness. Any one of the possible defects defined above shall be cause to consider

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a ballistics drive defective, unless separate defects per-hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Ballistics drives which have passed the above examinations shall be subjected to the tests of 4.6.3 to 4.6.3.5 inclusive to determine overall compliance with 3.3.3.

4.6.3 Performance.- The following tests shall be performed with the Ballistics Drives: M10, M10A1, M10A2, M10B1 and M10A6, positioned on the special testing equipment specified in 4.4.2.3.1. The Ballistics Drives: M10A3, M10A4 and M10A5 shall be positioned on the special testing equipment specified in 4.4.2.3.2. Setup of the test equipment and installation of the ballistics drive under test shall be performed in accordance with the setup procedures which are included on the test equipment drawing. Subsequent to installation, the ballistics drive shall be positioned and oriented in the following manner to conform to the requirements of 3.4.

(a) Sight through autocollimator "A" to the mirror block and autocollimate to the zero degree mirror by adjusting the trunnion arm of the test equipment. This is 0.0 degree gun elevation. Note the position of the level bubble in the level vial and the alinement of the zero graduation lines on the inner and outer scales of the double eccentrics. Failure of the bubble in the level vial to be centered within the graduation lines, or failure of the zero graduation lines on the inner and outer scales of the double eccentrics to be in coincidence with their respective index lines shall be cause for rejection of the ballistics drive under test.

(b) Apply a 200 inch pounds load on the cross shaft by mounting a 40 pound weight to the torque wheel assembly of the test equipment. With the stop pins retracted, attach the appropriate clamp block screw, or adapter assembly, to the superelevation input shaft of the ballistics drive under test.

(c) Rotate the input shaft in a counterclockwise direction, by means of the clamp block screw, or the adapter assembly, until the travel stop in the superelevation box is engaged. From this position rotate the input shaft in a clockwise direction the required number of turns indicated on the qualifying plate, minus one turn. Sight through autocollimator "B" to the mirror block "C". While viewing through the autocollimator, continue clockwise rotation of the input shaft until autocollimation is established to the zero degree mirror. Center the locating pin of the clamp block screw, or the adapter assembly, between the appropriate stop pins. This is 0.0 degree superelevation. Note the alinement of the index line on the input shaft with the arrow on the qualifying plate and record the complete number of clockwise turns the input shaft was rotated. Failure of the index line to be alined with the arrow in accordance with the value and direction indicated on the qualifying plate, shall be cause for rejection of the ballistics drive under test.

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4.6.3.1 Superelevation input travel.- Retract the stop pins of the clamp block screw, or the adapter assembly, to allow the input shaft to rotate freely during this test. Rotate the input shaft from its counterclockwise travel stop to its clockwise travel stop, then back to its counterclockwise travel stop. During this excursion, the number of turns of the input shaft shall be counted in one direction. Travel of the input shaft shall conform to the requirements of 3.4.1.1, as determined by a visual and tactile examination. Subsequent to this test, return the input shaft to 0.0 degree superelevation position, in accordance with the procedure specified in 4.6.3(c).

NOTE: The gun trunnion arm shall be immobilized at 0.0 degree gun elevation during this test (4.6.3.1) and for the following tests specified in 4.6.3.2 and 4.6.3.3.

4.6.3.2 Superelevation accuracy.- Retract the stop pins of the clamp block screw, or the adapter assembly. With backlash removed for clockwise travel, rotate the input shaft 5 complete revolutions clockwise from the 0.0 degree superelevation position, stopping at the fifth turn by extending the appropriate stop pin. Sight through autocollimator "B" to the appropriate mirror of mirror block "C". Record the position of the boresight point of the reflected image, as measured on the autocollimator. Retract the stop pin. Rotate the input shaft 5 additional complete revolutions clockwise (10 total from zero degree superelevation position), stopping at the tenth turn by extending the appropriate stop pin. Sight through autocollimator "B" to the appropriate mirror of mirror block "C". Record the position of the boresight point of the reflected image, as measured on the autocollimator. The rotational position of the output coupling, as determined by the recorded values, shall be within the tolerance specified in 3.4.1.2.

4.6.3.3 Superelevation backlash.- The testing equipment specified in 4.6.3 shall be utilized for this test, except the 40 pound weight mounted to the torque wheel assembly of the test equipment shall be removed. Perform this test in conjunction with 4.6.3.2. With backlash excluded for clockwise travel, position the input shaft at 0.0 degree superelevation, as specified in 4.6.3(c), and record this reading. Observe the index line on the input shaft and the arrow on the qualifying plate. The index line and the arrow on the qualifying plate will be misalined, due to removal of the 40 pound weight from the torque wheel assembly. Failure of the index line to be alined with the arrow shall not be cause for rejection when this test is performed. Retract the stop pins of the clamp block screw, or the adapter assembly. With backlash excluded, rotate the input shaft 1 additional turn clockwise, then reverse rotation of the input shaft (with backlash included) and stop at the zero turn position by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator. Retract the stop pin. With backlash excluded, rotate the input shaft 5 complete revolutions clockwise from the 0.0 degree superelevation position, stopping at the fifth turn by

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extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator. Retract the stop pin. With backlash excluded, rotate the input shaft 1 additional turn clockwise, then reverse rotation of the input shaft (with backlash included) and stop at the fifth turn position by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator. Again retract the stop pin. With backlash excluded, rotate the input shaft 5 additional complete revolutions clockwise (10 total from zero degree superelevation position), stopping at the tenth turn by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator reticle. Retract the stop pin. With backlash excluded, rotate the input shaft 1 additional turn clockwise, then reverse rotation of the input shaft (with backlash included) and stop at the tenth turn position by extending the appropriate stop pin. Sight through the autocollimator and record the position of the boresight point of the reflected image, as measured on the autocollimator. Any difference between the recorded readings at 0.0 degree superelevation, 5 complete clockwise revolutions, and 10 complete clockwise revolutions of the input shaft is the amount of superelevation backlash. Backlash resulting from rotation at any one of the three positions shall not exceed the limit specified in 3.4.1.2.1.

4.6.3.4 Synchronization accuracy.- This test shall be performed with the Shaft 8200797 or 8201837 disassembled from the Universal Assembly 8200862. This eliminates any planetary error due to superelevation input shaft rotation. The testing equipment specified in 4.6.3 shall be utilized for this test, except the mirror block "C" is not required. Mirror block "B" shall be mounted in position and used in place of mirror block "C". The gun trunnion arm shall be positioned at 0.0 degree gun elevation and the 40 pound weight shall be mounted to the torque wheel assembly of the test equipment. Sight through autocollimator "B" to the appropriate mirror of mirror block "B" and establish 0.0 degree superelevation, utilizing the procedure specified in 4.6.3(c). Without disturbing the 0.0 degree superelevation position, disassemble the shaft coupling from the coupling of the Universal Assembly by removing 4 screws. Prior to disassembly, the exact position of the Universal in relation to the assembly shall be established by appropriate reference marks, as it is necessary to return it to its original position when the specified tests have been completed. Subsequent to disassembly, sight through the autocollimator "B" and reestablish the 0.0 degree superelevation, providing autocollimation to the appropriate mirror of mirror block "B" has not been maintained. Do not disturb, nor make any further adjustments to the Universal Assembly for the remainder of the test. The following movements in elevation and depression shall be induced to the gun trunnion arm and autocollimator "A" by actuating the trunnion arm of the test equipment. Position the gun trunnion arm and autocollimator "A" at an angle greater than minus five degrees in depression. With backlash excluded, elevate the gun trunnion arm and autocollimator "A" until autocollimation is established

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to the minus five degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Elevate the gun trunnion arm and autocollimator "A" (without backlash) until autocollimation is established to the zero degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Elevate the gun trunnion arm and autocollimator "A" (without backlash) until autocollimation is established to the plus seventeen degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Elevate beyond the 17 degree mirror and eliminate backlash. Depress the gun trunnion arm and autocollimator "A" until autocollimation is established to the plus 17 degree mirror. Record this setting. Depress the gun trunnion arm and autocollimator (without backlash) until autocollimation is established to the zero degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image, as measured on the autocollimator. Depress the gun trunnion arm and autocollimator "A" (without backlash) until autocollimation is established to the minus five degree mirror. Sight through autocollimator "B" and record the position of the boresight point of the reflected image as measured on the autocollimator, At each of the recorded positions, the ballistics drive shall conform to the accuracy requirements of 3.4.2.1.

4.6.3.5 Synchronization, backlash.- Perform this test in conjunction with the test in 4.6.3.4. The difference in the two recorded readings (elevation and depression) at each position of zero, -5 and +17 degrees in 4.6.3.4 shall represent the amount of backlash resulting from change in direction during rotation of the input shaft. This difference shall not exceed the limit specified for backlash in 3.4.2.2.

4.6.4 Operability (60° to 90°F).- Prior to conducting the following operability tests, adjust the gun trunnion arm and autocollimator "A" until autocollimation is established to the zero degree mirror. While utilizing the reference marks made prior to disassembly, secure the coupling of the universal assembly and the shaft coupling in their original position.

4.6.4.1 Rotating studs.- Standard torque measuring equipment capable of measuring torque values of at least 5 pound inches shall be utilized for this test. An appropriate adapter, to accommodate the rotating studs, shall be used with the torque measuring equipment. Torque shall be measured while rotating the studs through their full range of travel and recorded for use in a subsequent test. The running torque required to rotate the studs shall not exceed the value specified in 3.4.3.1 for the temperature indicated.

4.6.4.2 Superelevation input shaft.- The torque measuring equipment specified in 4.6.4.1 shall be utilized for this test, except that an appropriate adapter to accommodate the superelevation input shaft

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(at its point of origin) shall be used. Torque shall be measured while rotating the superelevation input shaft throughout its entire range of travel clockwise and counterclockwise and recorded for use in a subsequent test. The running torque required to rotate the superelevation input shaft shall not exceed the value specified in 3.4.3.2 for the temperature indicated.

4.6.5 Operability (-40° and +150°F).

4.6.5.1 Rotating studs.- This test shall be performed by utilizing the torque measuring equipment specified in 4.6.4.1. The running torque required to rotate the studs shall not exceed the recorded value obtained in 4.6.4.1 by more than 1 pound inch at the temperature specified in 3.4.4.

4.6.5.2 Superelevation input shaft.- This test shall be performed by utilizing the torque measuring equipment specified in 4.6.4.1. The running torque required to rotate the superelevation input shaft (at its point of origin) shall not exceed the recorded value obtained in 4.6.4.2 by more than 1 pound inch at the temperatures specified in 3.4.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with the applicable packaging data sheet (see 2.1). The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The ballistics drives covered herein are differential drives which transmit superelevation data received from a ballistics computer to a rangefinder and gunner's periscope to provide superelevation for a given ammunition and range. Without change in superelevation, the ballistics drive acts as a solid linkage providing equal angular movement of the lines of sight of the rangefinder and gunner's periscope corresponding to the movement of the gun in elevation or depression.

6.1.1 The ballistics drives are intended for use in the fire control systems of the following tanks:

MODEL	Used with	Tank
M10	Periscope M31	M60
M10A1, or M10A6	Periscope M32 (Infrared)	M60
M10A2	Periscope M35 (Infrared)	M60
M10A3	Periscope M35 (Infrared)	M60A1
M10A4	Periscope M31	M60A1
M10A5	Periscope M32 (Infrared)	M60A1
M10B1	Periscope M31	M48A3

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Model of ballistics drive to be procured.
(c) Selection of applicable level of protection.
(d) Applicable packaging data sheet number (see 2.1 and 5.1).
(e) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A256

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